

Appendix A



Major Project 21CSA399A

Proposal

Title: Fake News Detection using Deep Learning

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Address of the Company:

Name, Designation and Communication details of the Guide:

(if the student chooses 'industry integration'):

Abstract

This project focuses on developing a Fake News Detection System using Deep Learning and Natural Language Processing (NLP) techniques. The system aims to classify news articles as fake or genuine based on textual content analysis.

The model will be built using the Fake News Dataset from Kaggle and implemented using Python, TensorFlow/Keras, and NLTK libraries. Long Short-Term Memory (LSTM) will be used to learn contextual relationships within news text.

The project involves text preprocessing, tokenization, word embedding, model training, and performance evaluation using accuracy, precision, recall, F1-score, and confusion matrix.

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Assumptions/Declarations:

- The dataset collected from Kaggle represents realistic fake and genuine news data.
- The project is developed for academic purposes only.
- Textual content contains sufficient information for classification.
- LSTM parameters will be optimized using standard evaluation methods.

Main Objective/Deliverable:

To design and implement a deep learning-based fake news detection system using LSTM and evaluate its performance using suitable classification metrics.

Deliverables:

- Trained deep learning model
- Python implementation with preprocessing and model training
- Performance analysis using accuracy, precision, recall, and F1-score
- Project documentation and result analysis

Timeline and Milestones:

	Milestones	Timeline
1.	Topic Selection and Proposal Submission	25/04/2026
2.	Literature Review and Background Study	26/04/2026 – 02/05/2026
3.	Dataset Collection and Understanding	03/05/2026 – 07/05/2026
4.	Text Preprocessing and Feature Extraction	08/05/2026 – 13/05/2026
5.	High-Level Design and Interim Report Preparation	18/05/2026
6.	LSTM Model Development and Training	19/05/2026 – 10/06/2026
7.	Model Evaluation and Optimization	11/06/2026 – 25/06/2026
8.	User Interface and Integration Development	26/06/2026 – 08/07/2026
9.	Final Documentation and Report Submission	18/07/2026
10.	Presentation and Demo Preparation	Before Final Review

Tools to be used.

Software/Hardware Tools	Specifications
Programming Language	Python 3.10+
Libraries	TensorFlow, Keras, Pandas, NumPy, NLTK, Matplotlib
Deep Learning Algorithm	LSTM
IDE / Environment	Visual Studio Code /Pycharm/ Jupyter Notebook
Dataset Source	Kaggle Fake News Dataset

Learning involved:

Topic	Description
Deep Learning Fundamentals	Understanding neural networks and sequential models.
Natural Language Processing	Learning text preprocessing and tokenization techniques.
Word Embeddings	Understanding text representation methods.
Model Evaluation	Learning accuracy, recall, precision, and F1-score interpretation.
Practical Deployment	Experience building a complete prediction system.

Date : 25/05/2026**Student Name and Signature: Harikrishnan R**